

Phytonutrient intake by adults in the United States in relation to fruit and vegetable consumption.



BACKGROUND: Individuals consuming diets dense in fruits and vegetables consume an array of phytonutrients as well as recognized nutritional components, including vitamins, minerals, and fiber. There is a growing body of evidence that phytonutrients may play positive roles in health. **OBJECTIVE:** The purpose of this research was to estimate usual intakes of nine individual phytonutrients by Americans consuming recommended levels of fruits and vegetables compared to intakes by adults not meeting these recommendations, and to identify contributions of food sources to total phytonutrient intakes. The phytonutrients examined in this study are found predominantly in fruits and vegetables. **DESIGN:** Food consumption data from the National Health and Nutrition Examination Surveys 2003-2006 and phytonutrient concentration data from US Department of Agriculture databases and the published literature were used to estimate energy-adjusted usual intakes. Student's t tests were used to compare mean energy-adjusted phytonutrient intakes between subpopulations who consumed recommended amounts of fruits and vegetables vs those who did not. Percentage contributions of each phytonutrient by food source were estimated for all adults. **RESULTS:** Energy-adjusted intakes of all phytonutrients other than ellagic acid were considerably higher among both men and women meeting dietary recommendations for fruit and vegetable intakes compared to those not meeting the recommendations; energy-adjusted intakes of ellagic acid were higher only among women meeting vs not meeting the recommendations. For five of the nine phytonutrients (α -carotene, β -cryptoxanthin, lycopene, hesperetin, and ellagic acid), a single food accounted for 64% or more of the total intake of the phytonutrient. **CONCLUSIONS:** Energy-adjusted intakes of carotenoids and flavonoids are higher among men and women whose diets conform to dietary guidance for fruits and vegetables. A limited number of foods provide the majority of these phytonutrients. Findings from this research provide important reference information on the phytonutrient contributions of a diet rich in fruits and vegetables.